

Islands Sate without Change in Backgroundedness

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Abstract Certain structural domains restrict syntactic movement, a phenomenon termed the island effect. Although traditionally considered a class of syntactic constraints, island effects have been attributed by some to the clash between the discourse backgrounded status of the island domain and the foregrounding function of movements. Although previous work supports a correlation between the discourse backgroundedness of structures and their islandhood, it is unclear whether island effects can be entirely reduced to a discourse constraint as some previous studies suggest. In this study, we provide evidence against this reductionist version of the discourse accounts of island. Specifically, using the island satiation effect as an experimental manipulation, we show that island effects ameliorate after repeated exposure, but there is no corresponding reduction in perceived backgroundedness of the island domains. This result demonstrates a dissociation between the satiation of island effects and any change in the perceived discourse backgroundedness of the island domain, challenging a core prediction of the discourse-only account of islands.

Keywords: Island effects; information structure; discourse backgroundedness; syntactic satiation; filler-gap dependency

1 Introduction

Languages allow long-distance dependencies between a moved constituent and its original position where it is interpreted, an example of which is the *wh*-filler-gap dependency in English resulting from *wh*-movements, as shown in (1) (Fodor 1978). The moved *wh*-element *what* appears in a sentence-initial position, but is interpreted in its original matrix object position marked with a gap.

- (1) What did the professor eat ___?

Such dependencies can be unbounded: there is in principle no limit on the number of words or clausal boundaries that a long-distance dependency can span. This is illustrated in (2).

- (2) a. What did the student claim that the professor ate ___?
b. What did the librarian hear that the student claimed that the professor ate ___?
c. What did the technician think that ... that the librarian heard that the student claimed that the professor ate ___?

Despite its general unboundedness, long-distance dependencies are constrained by certain structural domains, a phenomenon termed the *island effect* (Chomsky 1964; Ross 1967: *inter alia*). For example, as shown in (3) and (4), it is impossible to move a *wh*-

element from within a CP complement of N (the *complex-NP island effect*) or a subject DP (the *subject island effect*) in English.

- (3) The complex-NP island effect
 - a. The student made the claim that the professor ate a cake.
 - b. *What did the student make the claim that the professor ate ___?
- (4) The subject island effect
 - a. The owner thought that the dish by the famous chef would raise the restaurant reputation.
 - b. *Who did the owner think that the dish by ___ would raise the restaurant reputation?

Island effects have traditionally been analyzed as grammatical constraints on long-distance dependencies (Chomsky 1973; 1981; 1986; Ross 1967; Rizzi 1990; Bresnan 1976; Sag 1976: *inter alia*). Under this class of accounts, island effects arise as the result of violating certain representational or derivational constraints in the grammar. For example, Chomsky (1973) proposed the *Subjacency Condition* on transformation, which prohibits syntactic movements from crossing two or more *bounding nodes* (NP and TP in English), thereby capturing the ungrammaticality of sentences like (3)b and (4)b. In more recent minimalist theories of syntax, the Subjacency Condition is recapitulated as the *Phase Impenetrability Condition*, which states that syntactic derivations proceed phase-by-phase, and each phase (e.g., DP, vP, and CP in English), except for its edge, is opaque to further syntactic operations like movements once its derivation is complete (Chomsky 2000; 2001; 2004).

Alternatively, some studies argue that island effects do not reflect grammatical constraints but rather result from processing-level difficulties (Culicover et al. 2022; Hofmeister & Sag 2010; Hofmeister et al. 2013; Kluender 1991; Kluender & Kutas 1993; Deane 1991; Alexopoulou & Keller 2007). Under this class of accounts, island-violating sentences are grammatical but hard to process. Processing a long-distance dependency requires holding the filler (e.g., a *wh*-element) in working memory until the gap position is encountered. This process can be cognitively taxing, leading to perceived unacceptability in some cases. Nevertheless, there has been extensive evidence suggesting that even though processing-level factors do contribute to the degradedness of island-violating sentences, island effects cannot be entirely reduced to processing difficulties (Sprouse et al. 2012; Dillon & Hornstein 2013; Yoshida et al. 2014: *inter alia*).

Another line of work captures the island effects with discourse-level constraints (Erteschik-Shir 1973; Goldberg 2013; Cuneo & Goldberg 2022; 2023; Namboodiripad et al. 2022; Erteschik-Shir 2007; Abeillé et al. 2020; Winckel et al. 2025; Lu et al. 2025: *inter alia*). Erteschik-Shir (1973) first proposed that island effects can be captured by the interplay of the discourse properties of extractions and the concept of *semantic dominance*: a constituent is only extractable if the content it encodes may be interpreted as dominant. Erteschik-Shir (1973) further defines *dominance* based on Chomsky's (1969) definition for *focus* as non-presupposed constituents in sentences. In effect, Erteschik-Shir's constraint states that elements contained inside presupposed constituents cannot be extracted.

More recently, following Erteschik-Shir (1973), Goldberg (2006) proposed the Backgrounded Constructions are Islands (BCI) account (5).

- (5) The Backgrounded Constructions are Islands (BCI) account (Goldberg 2006: p.135)
Backgrounded constituents may not serve as gaps in filler-gap constructions.

In a series of experimental studies, Goldberg and colleagues provided support for the BCI account by showing that sentences with filler-gap dependencies are judged to be more acceptable when the extraction sites are interpreted as less discourse backgrounded. Operating on the assumption that backgrounded constituents are not affected by sentential negation whereas at-issue constituents are, Goldberg and colleagues operationalized *backgroundedness* with the negation task (Ambridge & Goldberg 2008; Goldberg 2013; Cuneo & Goldberg 2022; 2023). An example is shown in (6). In this example, if the information expressed by the clausal adjunct (that the owner gave Fido treats) is backgrounded, it should survive sentential negation. Therefore, the answer to the interpretation question should lean more towards “Yes” than “No”, and extraction from within the adjunct (example (7)) should be judged as degraded. When applied to a wide variety of island-violating sentences, the negation test’s results correlate with the extraction acceptability.

- (6) The negation task for testing constituent backgroundedness (Cuneo & Goldberg 2023).

Instruction: Please think about what the following sentence means.

The owner did not get Fido outside while giving him treats.

Question: Did the owner give Fido treats?

- (7) *What did the owner get Fido outside while giving him ___?

Lu et al. (2025) and Fergus et al. (2025) took a step beyond such correlational evidence, and further showed a causal relation between backgroundedness and islandhood: by focusing the extracted element through changing the prosodic contours of sentences, certain islands (manner-of-speaking islands and conjunct islands) can be ameliorated. Yet, neither showed that island effects can be completely voided. This leaves open a possibility: although the discourse status of a constituent affects its islandhood, non-discourse factors (e.g., constraints in syntax) may also be at play. Can island effects be *completely* reduced to a single discourse constraint such as the BCI? The current study aims to answer this question.

In this study, we make use of the exposure-induced amelioration of island violation (i.e., the *island satiation* effect), and test whether the increase in the acceptability of island-violating sentences necessarily reflects a change in the participants’ information structural perception of the island domains, as predicted by a discourse-only account of islands. The rest of the paper is organized as follows: in Section 2, we will introduce the island satiation effect. In Sections 3 and 4, we present two experimental studies that test whether the satiation of complex-NP and subject islands reflects underlying changes in the discourse statuses of the island domains. In Section 5, we provide a general discussion of the findings and the implications for the theories of islands. Section 6 concludes the paper.

2 The island satiation effect

Sentence acceptability is not invariant. It has been experimentally shown that after being repeatedly exposed to certain degraded sentence types, comprehenders tend to rate such sentences as increasingly more acceptable (the *satiation effect*: Snyder 2000; 2022b: *inter alia*). Various types of island-violating sentences are affected by satiation, including sentences like (3)b and (4)b that violate the complex-NP island constraint (Snyder 2000; Hofmeister & Sag 2010; Do & Kaiser 2017; Lu et al. 2021; Snyder 2022a) or the subject

island constraint (Francom 2009; Chaves & Dery 2014; 2019; Chaves & Putnam 2020; Chaves et al. 2024; Lu et al. 2021; 2022).¹

In the present study, we aim to use island satiation as an experimental manipulation to test the reductionist version of the discourse accounts of islands. Under such accounts, islandhood is reduced to the discourse backgrounded or presupposed status of the structural domains. Therefore, any change in the acceptability of island violations should reflect an underlying change in the discourse status of the relevant island domain. The goal of the present study is to test this prediction. Specifically, we use island satiation to reduce the magnitude of island effects, and compare the participants' interpretations of the information structural status of the island domains pre- and post-satiation. This experimental design provides a testing ground for the discourse account: if islandhood can be completely reduced to information structure, the improvement in acceptability of island violations should reflect a reduction in the degree to which the island domain is perceived as discourse backgrounded.

For this design to work, we need to ensure that island satiation reflects a reduction in the island violation itself, rather than a general increase in the acceptability of island-violating sentences. This distinction is crucial because the acceptability increase of an island-violating sentence may be driven by factors that are orthogonal to the island constraint itself. For example, it has been claimed that repeated exposure facilitates the processing of long-distance dependencies (Francom 2009; Hofmeister & Sag 2010). Under this view, comprehenders are able to re-allocate their working memory resources to better process sentences with long-distance dependencies (including but not limited to island-violating sentences). The absolute ratings of island-violating sentences may increase simply because participants become better at processing long-distance dependencies in general. Therefore, for the present study, it is not sufficient to show that island-violating sentences receive higher ratings after exposure. We must instead show that the constraint itself weakens, that is to say, the island effect itself decreases in magnitude over the course of exposure.

To isolate the satiation of the island effect from other factors that may also influence the acceptability of island-violating sentences, we adopt the experimental design of Lu et al. (2026b), which combines a repeated exposure paradigm with the factorial probe for island effects introduced by Sprouse et al. (2012). In the factorial design, island effects are defined using a 2×2 design crossing two factors: *Structure* (Island vs. Non-Island) and *Length* (Embedded: the dependency spans the embedded structural boundary vs. Matrix: the dependency is in the matrix clause). Example sentences of the four conditions from the factorial design are shown in (8).

(8) Example factorial design (Sprouse et al. 2012).

- a. ISLAND, EMBEDDED
What did you make the claim that John bought ___?
- b. ISLAND, MATRIX
Who ___ made the claim that John bought a car?
- c. NON-ISLAND, EMBEDDED
What did you claim that John bought ___?

¹ There have also been studies reporting no satiation for subject island and complex-NP islands (Sprouse 2009; Hiramatsu 2001). See Snyder (2022b) and Lu et al. (2024) for overviews of empirical findings on island satiation. In particular, Lu et al. (2024) presented meta-analytic evidence suggesting that both subject island and complex-NP island demonstrate satiation, and the null results reported in the literature are likely due to the lack of power of the design of early studies on satiation.

d. NON-ISLAND, MATRIX

Who ___ claimed that John bought a car?

Both island structure and cross-clausal dependencies may independently reduce sentence acceptability: longer dependencies are more costly to process than shorter ones, and island structures (e.g., complex-NP structures, relative clauses, etc.) may be less frequent than non-island structures (e.g., complement clauses) and are thus less acceptable. The island effect, which arises only when a long-distance dependency crosses an island structural boundary, is then probed as an interaction of the two factors (i.e., the super-additive penalty on acceptability when both an island structure and a long dependency crossing the structure are present simultaneously).

The factorial design identifies the super-additive penalty associated with extraction from an island domain, but does not by itself determine the source of this penalty. The interaction establishes that the degradation cannot be reduced to the additive effects of dependency length and island structure alone, but it cannot inform whether the additional penalty is syntactic, discourse-based, processing-related, or some combination of these factors (Kim 2021). In the present study, we use the super-additive island effect only as an operational measure of the “island penalty” and ask whether its reduction over repeated exposure is accompanied by a corresponding change in the backgroundedness of the island domain.

By combining the factorial design with the repeated exposure paradigm, we can further define island satiation as a reduction in the magnitude of the Structure $\times$ Length interaction over the course of the experiment. Statistically, this corresponds to a significant three-way interaction among Structure, Length, and Block, such that the super-additive island effect decreases across blocks. This design allows us to determine whether repeated exposure specifically weakens the island effect itself, rather than merely increasing the acceptability of island-violating sentences for reasons unrelated to islandhood.

Having established a way to measure the satiation of the island effect itself, we can use island satiation as a manipulation to test a central prediction of the reductionist version of the discourse accounts of islands. If island effects are fully reducible to the discourse status of the island domain, then a change in islandhood should reflect a change in the discourse status of the island domain. Specifically, when the Structure $\times$ Length interaction decreases over exposure, the island domain should also be interpreted as less discourse backgrounded after exposure. The following experiments test this prediction for complex-NP islands and subject islands.

3 Experiment 1

Experiment 1 tests whether the amelioration of the complex-NP island effect through satiation reflects an underlying change in the perceived discourse backgroundedness of the complex-NP island domain. We use a three-phase experimental design, including a pre-exposure phase, an exposure phase, and a post-exposure phase. In the pre-exposure phase, participants complete a negation task to measure their initial perception of the backgrounded status of the complex-NP island domain. Then, entering the exposure phase, participants complete an acceptability-rating task in 4 repetitions, designed to induce satiation of the complex-NP island effect. Lastly, in the post-exposure phase, participants complete a second negation task to measure their final perception of the backgrounded status of the complex-NP island domain.

If the complex-NP island effect satiates, we expect the magnitude of the Structure $\times$ Length interaction to decrease over exposure blocks. If this reduction in islandhood is

accompanied by a corresponding reduction in the perceived backgroundedness degree of the complex-NP domain, this would suggest that the amelioration of the complex-NP island effect is associated with a change in the information-structural status of the island domain, consistent with the prediction of the discourse account of islands. Conversely, if the complex-NP island effect decreases over exposure without a reduction in perceived backgroundedness, this would suggest that island satiation can occur independently of changes in information-structural interpretation, challenging the prediction of the discourse account of islands.

3.1 Participants

A total of 300 native English speakers were recruited on *Prolific*, of whom 6 were excluded. Participants were excluded if they answered more than one practice trial incorrectly or if the 95% confidence intervals for their mean responses to grammatical and ungrammatical fillers overlapped.

3.2 Materials and Procedures

In Experiment 1, we used a three-phase experimental design, including a pre-exposure phase, an exposure phase, and a post-exposure phase. In the pre- and post-exposure phases, we used the negation task to probe the backgroundedness of the island domains. In the exposure phase, we used an acceptability rating task with a 2×2 factorial design following Sprouse et al. (2012), manipulating *Structure* (island vs. non-island) and *Length* (short dependency vs. long dependency) to probe the satiation of the complex-NP island effect.

We constructed 24 critical lexicalizations, each with variants for both the negation tasks and acceptability rating tasks. For each participant, the 24 lexicalizations were randomly divided across the 3 phases: 4 appeared in the pre-exposure negation task, 16 in the exposure phase, and 4 in the post-exposure negation task. Lexicalizations did not overlap across phases, to prevent repeated lexical content from introducing familiarity.

The pre- and post-exposure phase each contains 4 critical items and 8 fillers. The exposure phase consists of 4 unannounced blocks, each containing 4 critical sentences (1 per condition) and 8 fillers.

3.2.1 Negation task

Following previous work, we probed the degree of discourse backgroundedness of the complex-NP island domain using the negation task (Ambridge & Goldberg 2008; Cuneo & Goldberg 2023; 2022; Namboodiripad et al. 2022: *inter alia*). Participants read a declarative context sentence containing an island domain and then answer an interpretation question targeting that island. For the context sentences, each item appeared in one of the two polarity conditions: the Baseline non-negated condition, or the Negated condition which contains a matrix sentential negation. For the interpretation questions, we provided continuous slider scales that record responses ranging from 0 “Definitely No” to 1 “Definitely Yes”. See sample stimuli in (9).

- (9) Instruction: Please think about what the following sentence means.
Context sentence:
(Baseline Condition) David discussed the claim that the airline canceled the morning flight to Chicago.

(Negated Condition) David didn't discuss the claim that the airline canceled the morning flight to Chicago.

Question: Was there a claim that the airline canceled the morning flight to Chicago?

We took the difference in responses between Baseline and Negated conditions to index the degree of backgroundedness of the island domain. If the questioned constituent is foregrounded, the difference is expected to be large. For instance, if the question to the two context sentences in ((9)) targets the at-issue matrix predicate "Did David discuss the claim?", the response to the Baseline Condition will be closer to "Definitely Yes" and the response to the Negated Condition would be closer to "Definitely No", resulting in a large contrast. By contrast, if the questioned constituent is backgrounded, the difference in responses to the two conditions is expected to be minimal.

3.2.2 Acceptability rating task

Following Sprouse et al. (2012) among others, we used a 2×2 factorial design crossing *Structure* and *Length* to probe island effects. We provided continuous slider scales that record participant responses ranging from 0 "Totally Unacceptable" to 1 "Totally Acceptable".² The magnitude of the island effect can be quantified as a Difference-of-Differences (DD) score, which is calculated by first taking the difference between the island and non-island conditions at each level of *Length*, and then taking the difference between these 2 differences. A larger positive DD score corresponds to a stronger island effect. Remaining statistically equivalent, the island magnitude can also be quantified as the super-additive interaction between *Structure* and *Length*. See sample stimuli in (10).

- (10)
- a. ISLAND, EMBEDDED
What did David discuss the claim that the airline canceled ___?
 - b. ISLAND, MATRIX
Who ___ discussed the claim that the airline canceled the morning flight to Chicago?
 - c. NON-ISLAND, EMBEDDED
What did David claim that the airline canceled ___?
 - d. NON-ISLAND, MATRIX
Who ___ claims that the airline canceled the morning flight to Chicago?

If islands satiate, we expect island effects to decrease after exposure, demonstrated by a significant *Structure* $\times$ *Length* $\times$ *Block* interaction such that the super-additive *Structure* $\times$ *Length* interaction decreases in magnitude as block increases. If the discourse account is correct in predicting that the change in perceived backgroundedness is the underlying driving factor for the change in island-violation acceptability, we expect the degree of backgroundedness of the island domains to decrease after exposure, demonstrated by a significant *Polarity* $\times$ *Task Position* interaction.

² Although Likert scales have been the norm in early experiment syntax studies (e.g., Sprouse et al. 2012), continuous slider scales have been widely adopted in more recent studies on island effects and island satiation (Coopmans et al. 2025; Lu et al. 2021; 2022), as well as acceptability judgment experiments testing other phenomena (Ambridge et al. 2026; Cremers & Chemla 2017; Lu et al. 2025; 2026a; Schoenmakers 2023; Schoenmakers & Van Hout 2024; *inter alia*), and therefore do not constitute a novel methodology. Moreover, in a large-scale meta-analysis of island satiation studies, Lu et al. (2024) explicitly tested whether scale type affects the magnitude of island satiation in a moderator analysis, and found that scale type is not a significant moderator of satiation. Given the established use of continuous scales in the satiation literature and the lack of evidence that scale type systematically modulates satiation, we think the current use of a slider scale is well-suited for interpreting the present results.

3.3 Results and Discussion

Acceptability rating results are shown in Figure 1. Negation task results are shown in Figure 2.

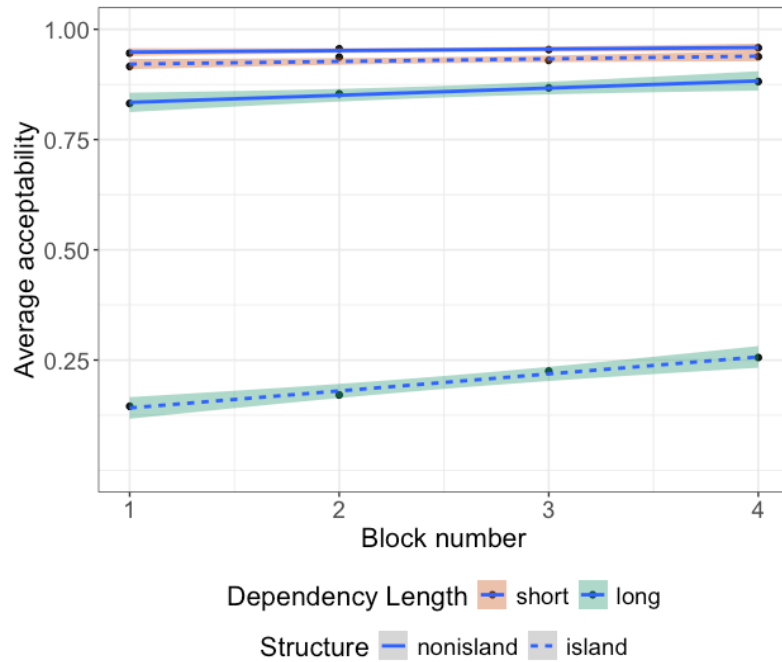


Figure 1: Acceptability ratings for Exp.1

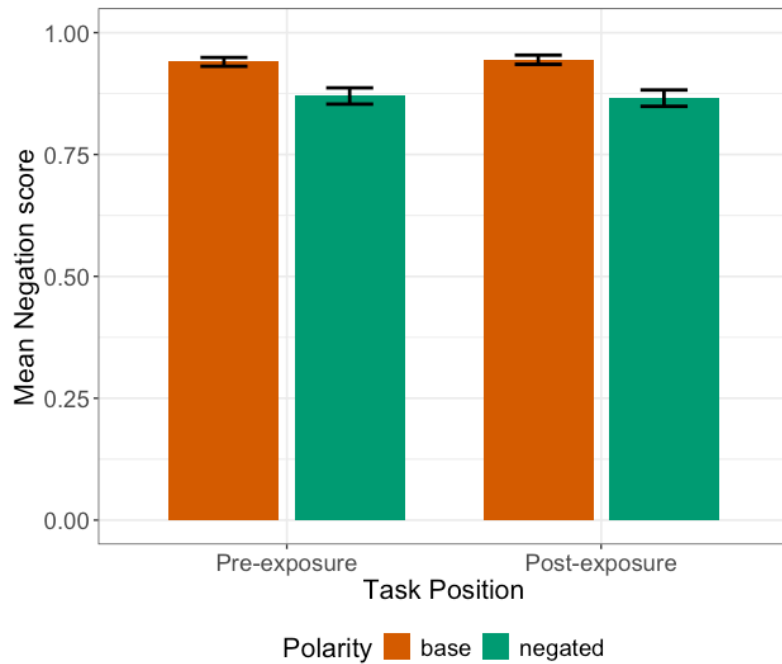


Figure 2: Negation task scores for Exp.1

We fitted a linear mixed-effects regression model predicting acceptability ratings with sum-coded fixed effects of Structure, Length, Block, and their interactions, and the maximal by-participant and by-item random effect structure allowing convergence. A significant Structure $\times$ Length interaction ($\beta = -0.44$, $SE = 0.006$, $t = -68.74$, $p < 0.001$) confirms a complex-NP island effect. We also found a significant three-way interaction ($\beta = 0.021$, $SE = 0.006$, $t = 3.72$, $p < 0.001$), which further indicates that the island effect decreased over blocks.

We further fitted a linear mixed-effects regression model predicting the negation task scores with sum-coded fixed effects of Polarity, Task Position, and their interaction, and the maximal by-participant and by-item random effect structure. There is no effect of Task Position ($\beta = 0.000016$, $SE = 0.0036$, $t = 0.005$, $p = 0.99$). There is a significant Polarity effect ($\beta = -0.037$, $SE = 0.005$, $t = 7.46$, $p < 0.001$) such that the Negated Condition received lower negation score than the Baseline Condition. Crucially, there is no Polarity $\times$ Task Position interaction ($\beta = -0.0021$, $SE = 0.0029$, $t = -0.72$, $p = 0.47$), indicating that backgroundedness of the complex-NP domain did not change across phases.

To verify that the negation task results truly reflect a null effect, we further computed the Bayes factors for the null Polarity $\times$ Task Position interaction effect using the BayesFactor package (Morey et al. 2015) in R, with medium ($r = 0.707$), wide ($r = 1$) and ultrawide ($r = 1.414$) Cauchy priors on effect size. We found moderate to strong evidence for the null hypothesis ($BF_{10} = 0.071$ under the medium prior, $BF_{10} = 0.056$ under the wide prior, and $BF_{10} = 0.038$ under the ultrawide prior), indicating that the observed data were substantially more likely under a model without a pre- to post-exposure change in backgroundedness.

In sum, in Experiment 1, we did find a significant island satiation effect (i.e., the complex-NP island magnitude decreased significantly throughout exposure), yet there is no change in the discourse status of the complex-NP island domain. These results challenge a discourse-only account of the complex-NP island.

4 Experiment 2

Experiment 2 applies the same design as Experiment 1 to the subject island effect. Specifically, Experiment 2 tests whether the amelioration of the subject island effect through satiation reflects an underlying change in the perceived discourse backgroundedness of the subject island domain.

4.1 Participants

A total of 300 native English speakers were recruited on *Prolific*, 15 were excluded on the basis of the same exclusion criteria as in Experiment 1.

4.2 Materials and Procedures

Experiment 2 shares the same three-phase experimental design with Experiment 1, including a pre-exposure phase, an exposure phase, and a post-exposure phase. In the pre- and post-exposure phases, we used the same negation task to probe the backgroundedness of the island domains. See sample stimuli in (11).

- (11) Instruction: Please think about what the following sentence means.
Context sentence:

(Baseline Condition) The programmer thought the software update by the lead developer would fix the security issue.

(Negated Condition) The programmer didn't think the software update by the lead developer would fix the security issue.

Question: Was the software update by the lead developer?

In the exposure phase, we employed the same acceptability rating task to probe the super-additive island effects. See sample stimuli in (12).

- (12)
- a. ISLAND, EMBEDDED
Who did the programmer think the software update by ___ would fix the security issue?
 - b. ISLAND, MATRIX
Who ___ thought the software update by the lead developer would fix the security issue?
 - c. NON-ISLAND, EMBEDDED
Who did the programmer think ___ would fix the security issue?
 - d. NON-ISLAND, MATRIX
Who ___ thought the lead developer would fix the security issue?

4.3 Results and Discussion

Acceptability rating results are shown in Figure 3. Negation task results are shown in Figure 4.

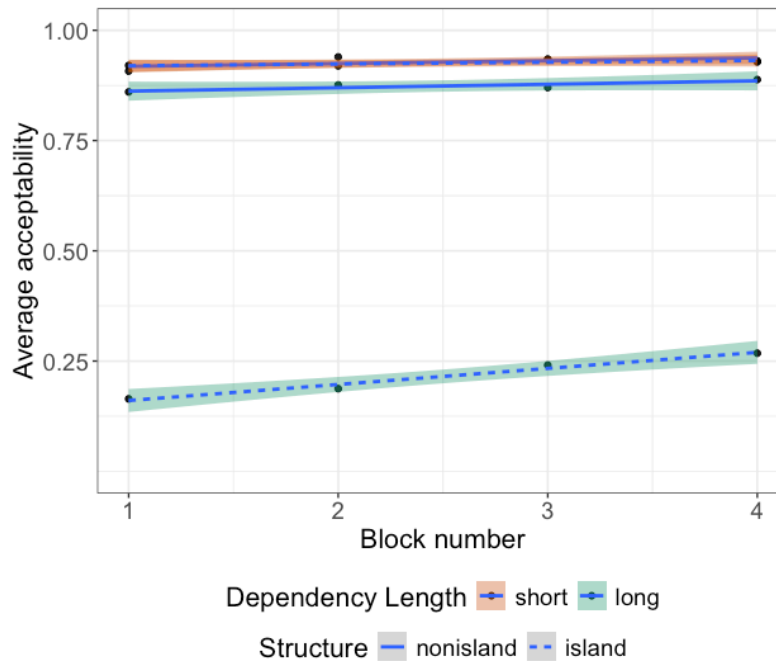


Figure 3: Acceptability ratings for Exp.2

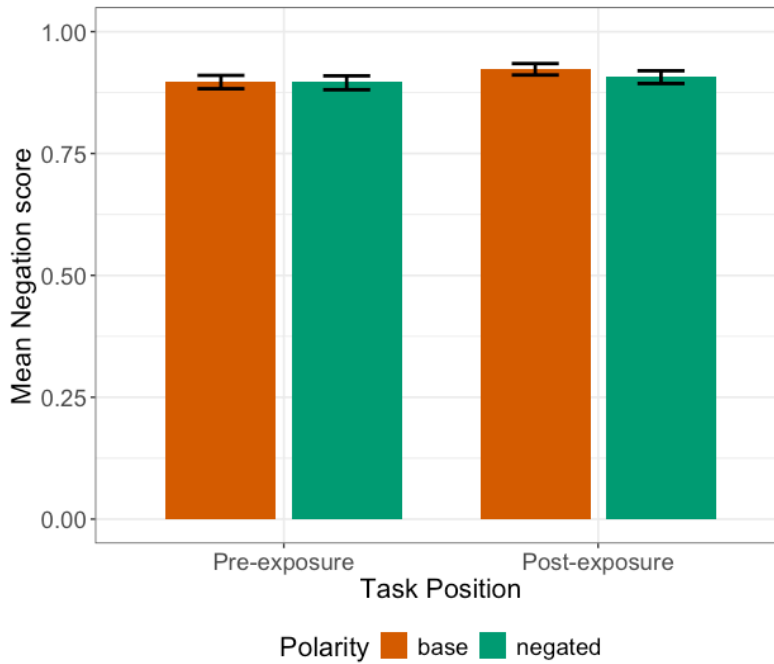


Figure 4: Negation task scores for Exp.2

We fitted a linear mixed-effects regression model predicting acceptability ratings with sum-coded fixed effects of Structure, Length, Block, and their interactions, and the maximal by-participant and by-item random effect structure allowing convergence. A significant Structure $\times$ Length interaction ($\beta = -0.16$, $SE = 0.005$, $t = -32.86$, $p < 0.001$) confirms a subject island effect. A significant three-way interaction ($\beta = 0.008$, $SE = 0.003$, $t = 3.33$, $p < 0.01$) further indicates that the island effect decreased over blocks, again suggesting satiation.

We further fitted a linear mixed-effects regression model predicting the negation task scores with sum-coded fixed effects of Polarity, Task Position, and their interaction, and the maximal by-participant and by-item random effect structure. There was a significant effect of Task Position ($\beta = -0.0094$, $SE = 0.0037$, $t = -2.54$, $p < 0.05$) where the post-exposure negation score is higher than the pre-exposure scores. There is no significant main effect of Polarity ($\beta = 0.0044$, $SE = 0.0033$, $t = 1.37$, $p = 0.18$). Crucially, there is no Polarity $\times$ Task Position interaction effect ($\beta = -0.003$, $SE = 0.003$, $t = -1.00$, $p = 0.32$), indicating that backgroundedness of the subject island domain did not change across phases.

To verify the negation task results truly reflect a null effect, we further computed the Bayes factors for the null Polarity $\times$ Task Position interaction effect using the BayesFactor package (Morey et al. 2015) in R, with medium ($r = 0.707$), wide ($r = 1$) and ultrawide ($r = 1.414$) Cauchy priors on effect size. We found moderate to strong evidence for the null hypothesis ($BF_{10} = 0.10$ under the medium prior, $BF_{10} = 0.080$ under the wide prior, and $BF_{10} = 0.054$ under the ultrawide prior), indicating that the observed data were substantially more likely under a model without a pre- to post-exposure change in backgroundedness.

One anonymous reviewer pointed out that the PPs headed by *by* inside the complex subjects, which is present in all 24 critical items of the current experiment, could independently contribute to acceptability degradation, regardless of extraction. We acknowledge that this is possible. However, both the [island, long] and the [island, short] conditions

contain such PPs, and any effect on acceptability by such PPs should arise as a part of the main effect of *Structure*, and should not affect the interaction of *Structure* and *Length* which we used as the measure for the island effect. To further address the concern, we conducted a follow-up experiment testing the acceptability of the declarative forms of the original stimuli sentences, along with the same filler sentences as in Exp.2. Example stimuli of the follow-up norming experiment are shown in (13). We collected data from 96 self-reported monolingual English speakers through Prolific. The results show that the *by*-PP sentences (the declarative version of the [island, long/short] conditions in Exp.2) received a mean rating of 0.907 (95% CI: [0.898, 0.915]), and the simple subject sentences (the declarative version of the [non-island, long/short] conditions in Exp.2) received a mean rating of 0.921 (95% CI: [0.913, 0.929]). The acceptability contrast, which is less than 0.02 on a 0–1 scale, is minimal. Therefore, we believe that the use of PPs headed by *by* in the original stimuli is unlikely to affect the overall acceptability pattern.

(13) Example stimuli of the follow-up norming experiment

- a. BY-PP SUBJECT: The programmer thought the software update by the lead programmer would fix the security issue.
- b. SIMPLE SUBJECT: The programmer thought the lead programmer would fix the security issue.

In sum, the results of Experiment 2 point to a significant island satiation effect (i.e., the subject island magnitude decreased significantly), yet there is no change in the discourse status of the subject domain. These results challenge a discourse-only account of the subject island effect.

5 Discussion

5.1 Summary of Findings

The present study examines whether the amelioration of island effects through satiation reflects a corresponding change in the perceived discourse backgroundedness of the island domain. In two experiments, we found that both complex-NP islands and subject islands exhibit reliable satiation: the magnitude of the super-additive *Structure* × *Length* interaction decreased significantly over exposure. At the same time, there was no corresponding reduction in the degree to which the island domains were perceived as discourse backgrounded by the participants. Bayes factor analyses further showed strong evidence for the lack of change in the perceived discourse backgroundedness for both experiments.

5.2 Implications for Theories of Islands

The dissociation between satiation and changes in discourse backgroundedness challenges a central prediction of the reductionist version of the discourse accounts of islands. Under such accounts which attribute islandhood entirely to the discourse backgroundedness of the island domain, any reduction in the perceived island effect should be accompanied by a corresponding reduction in perceived backgroundedness. This prediction was not borne out in either experiment. Complex-NP and subject island effects weakened over exposure, while perceived backgroundedness remained stable.

Our findings are compatible with an analysis in which complex-NP and subject islands are (at least partially) attributed to syntactic constraints. Through repeated exposure,

the comprehenders “unlearn” the constraints that are responsible for the initial degradation of the island-violating sentences. The exact mechanism of how a syntactic island constraint can be unlearned is not central to the interpretation of our results, but below we offer a possibility. One possible learning algorithm is proposed in [Pearl & Sprouse \(2013\)](#), in which the acquisition of *wh*-movement is modeled as the distributional learning of possible sequences of maximal projections that can be spanned by movement, and island effects emerge when certain projection sequences are rarely spanned by movements in a learner’s input.³ Under this view, throughout repeated exposure to island-violating sentences, the comprehenders acquire the maximal projection sequences involved in complex-NP and subject islands as possible movement paths. The island effects thus show amelioration. In this process, the comprehenders’ syntactic knowledge (the knowledge of whether a projection sequence forms a valid movement path) is updated through learning, but the information structural interpretation of the island domain remains constant.

Another alternative is that the satiating component of the island penalty reflects the relative frequency of island-violating dependencies rather than a categorical syntactic constraint. If island-violating dependencies are relatively rare in the input, repeated exposure may increase their familiarity and reduce the super-additive penalty without changing the discourse status of the island domain.

Alternatively, the observed dissociation may arise if repeated exposure changes the participants’ tolerance of extraction from backgrounded domains without changing the information-structural status of the domains themselves. Under this account, participants may continue to perceive an island domain as backgrounded, while learning that a filler-gap dependency targeting backgrounded information can nevertheless be accommodated. It is worth noting that this possibility would require independent evidence that exposure specifically changes the tolerance of *wh*-extraction from backgrounded information, and we leave it open for future work.

More generally, our results do not imply that discourse factors play no role in island effects. The satiation effects we observed in the two experiments do not entirely wipe away the complex-NP and subject island penalties. It is possible that island effects are not monolithic, and the violation of discourse-level constraints still accounts for a proportion of the invariant component of island effects. This is in line with recent findings that through foregrounding elements inside island domains, island effects can be partially ameliorated but not entirely erased ([Lu et al. 2025](#); [Fergus et al. 2025](#)). Taken together, these findings suggest that island effects (at least the complex-NP island and subject island that were tested in the current study) likely arise from both grammatical constraints and discourse constraints.⁴

5.3 Implications for Theories of Satiation

Finally, although the present study was not designed to adjudicate among the competing theories of island satiation, the results nevertheless carry implications about satiation. In the current study, we operationalized island satiation as a reduction in the super-additive Structure $\times$ Length interaction over exposure. Therefore, the observed effects specifically

³ See [Legate & Yang \(2025\)](#) for a similar proposal, in which long-distance movements are formulated as chains of strictly local steps (from the specifier of one projection to the specifier of the immediately dominating projection), and the acquisition of movements is formulated as the learning of what local movement steps are allowed.

⁴ See [Haegeman et al. \(2014\)](#) for a proposal in which subject island effects arise from the cumulative violation of multiple constraints rather than from a single categorical constraint.

reflect amelioration of the island constraint itself, rather than changes in processing-level sources of unacceptability such as the penalties on long-distance dependencies and low-frequency structures, which are orthogonal to the island constraint itself. Our results suggest that island satiation cannot be explained solely as a general improvement in processing, contrary to claims made in some previous studies on island satiation (The memory-bottleneck account of satiation: Snyder 2000; Francom 2009; Hofmeister & Sag 2010: *inter alia*).

Task-based accounts of satiation are also unlikely to explain the results of the current study. For example, Sprouse (2009) suggested that satiation may arise from an equalization response strategy, whereby participants try to balance their positive and negative responses on the acceptability scale. When an experiment contains a predominance of degraded sentences, participants may therefore give increasingly positive ratings to balance their overall responses. Our experiments contained equal numbers of grammatical and ungrammatical fillers, thereby limiting the imbalance among the filler sentences. The critical sentences are overwhelmingly acceptable: for both experiments, three out of the four critical conditions (all except the [island, long] condition) in the factorial design received average ratings above 0.75 on the 0–1 acceptability scale. Even if the participants adopted the equalization strategy during the experiment, the strategy would drive the overall ratings of sentences downwards as opposed to upwards on the acceptability scale. Another influential task-based account of satiation was proposed by Stepanov (2024) and Novozhilov et al. (2025), based on the findings by Brown et al. (2021) that satiation mostly affects sentence types with “middle-of-the-scale” ratings. Under this proposal, sentence types that satiate are grammatical, yet receive middle-of-the-scale ratings because participants are uncertain about their grammaticality status; with repeated exposure, participants gain confidence in their judgments, and thus raise the ratings for such sentences. However, the island-violating sentences in both experiments of the current study received initial acceptability ratings below 0.25 on the 0–1 acceptability scale, which is well outside the “middle-of-the-scale” range identified by Brown et al. (2021) (2.75 to 5.25 on a 1–7 scale, which maps to 0.3 to 0.7 on a 0–1 scale). Therefore, the present results do not straightforwardly follow either of these task-based accounts.

5.4 Limitations

One limitation of the current study is that any implication we may draw from the results is crucially limited to the particular island configurations we tested. We only examined two island types: complex-NP island, and subject island. Furthermore, we only tested two specific forms of these two island types: complex-NP island with NP *claim*, and subject island with definite DP subjects containing PP complements to the head noun. There has been documented variation in a DP’s opacity to extraction dependent on the lexical identity of the head noun (Ross 1967), and the definiteness, uniqueness, and specificity of the DP (Fiengo & Higginbotham 1981; Shen & Huang 2026; Tollan et al. 2025; Davies & Dubinsky 2003; Huang 2018: *inter alia*).

Another limitation of the current study involves the interpretation of null results. The conclusion of the current study rests on the null results from the backgroundedness tasks in the two experiments. The Bayes Factor analyses found strong evidence for the lack of effects in the backgroundedness tasks, and we also believe that the sample size ($N = 300$ for both experiments) is reasonably large, yet it is still in principle possible that we failed to detect an effect due to the lack of power. We leave it to future studies to investigate this possibility by conducting larger-scale experiments on this matter.

6 Conclusion

Our results show a clear dissociation between island satiation and change in discourse backgroundedness. While the complex-NP and subject island effect ameliorated with repeated exposure, there was no corresponding shift in the perceived backgroundedness of the relevant island domain by the participants. These results suggest that island satiation is not coupled with changes in discourse status, challenging a core prediction of the discourse-only accounts of island effects. Nevertheless, the findings leave open the possibility that island effects are partially, though not entirely, attributable to discourse factors.

Data Availability Statement

All data files and analysis scripts of the current project are available at <https://osf.io/e52tf/>.

Ethics and Consent

The experiments included in this study have been approved by the Northwestern University Institutional Review Board, approval number STU00224884.

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Competing interests

The authors have no competing interests to declare.

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